

Industrial Safety

Personal Protection Equipment (PPE)



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PPE

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PPE

1. Personal Protection Equipment PPE

PPE's are used to protect workers from injuries – by reducing the severity of injury in the event of an accident.

PPE's provided to workers should meet the relevant IS standards.

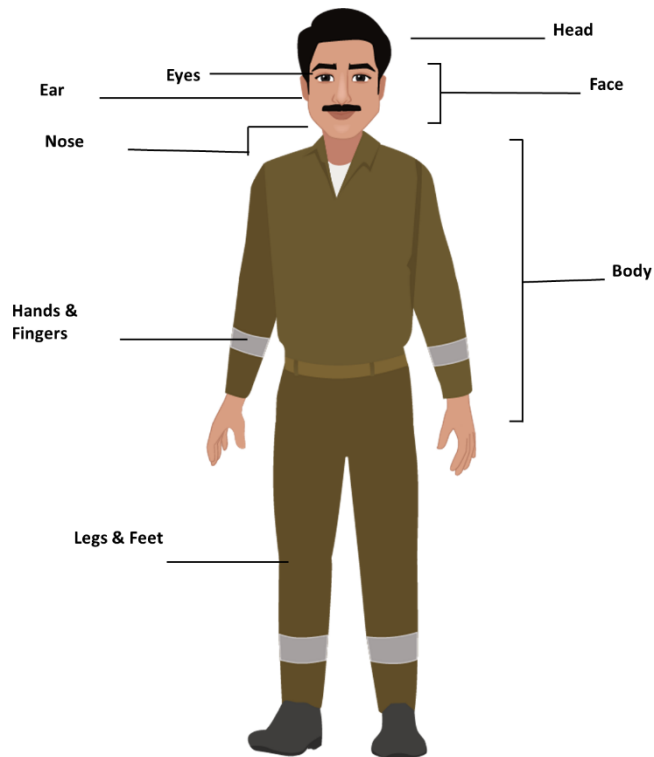


PPE

2. Body Protection using PPE

- ▶ Head
- ▶ Eyes
- ▶ Face
- ▶ Ear
- ▶ Nose (Respiratory)
- ▶ Hands & Fingers
- ▶ Legs & Feet
- ▶ Body (Visibility, Fall)

Specialized PPE are sometimes required for Chemical, Biological, Electrical, Heat, Fire protection.



PPE

2.1 Head Protection

Head injuries are one of the most common causes of fatality.

Safety helmets are hard hats designed to protect the head from:

- ▶ Impact from falling & flying objects
- ▶ Electric shocks & burns
- ▶ Overhead spills
- ▶ Exposure to dust
- ▶ Hair from getting caught in machinery



Hard Hat



PPE

2.2 Eye Protection

The Eyes are to be protected from

- ▶ Bright flashes
- ▶ Flying objects
- ▶ Dust
- ▶ Chemical splashes
- ▶ Gas and vapour
- ▶ Radiation

PPE include Safety glasses and goggles.



Safety Glasses



Safety Goggles



PPE

2.3 Face Protection

The Face including the eyes are to be protected from

- ▶ Bright flashes
- ▶ Flying objects
- ▶ Dust
- ▶ Chemical splashes
- ▶ Gas and vapour
- ▶ Radiation

PPE include wearable and handheld Face shields.



Face Shield



**Handheld
Face Shield**



2.4 Ear Protection

Excessive exposure to high levels of noise cause damage to hearing and lead to other illness.

PPE to protect against noise include Ear plugs and Ear muffs



Ear Plugs



Ear muffs



2.5 Nose (Respiratory) Protection

Protection is needed against inhalation of dust, gas and vapour.

Respirators serve to protect from breathing in such contaminants and include Disposable respirators, Particle Respirators and Gas Masks.



**N95 disposable
respirator**



Particle Respirator



Gas mask



PPE

2.6 Hand Protection

Hands, Fingers and Arms need protection against

- ▶ Impacts, Cuts, Punctures, Abrasion
- ▶ Chemical contact
- ▶ Electrical shocks
- ▶ Exposure to dust, dirt, germs

Hand protection PPE include various types of gloves, thumb and finger guards and Sleeves.



PPE

2.7 Leg Protection

Legs need protection against:

- ▶ Collusion against hard & sharp objects
- ▶ Wet slippery surfaces
- ▶ Falling and rolling objects
- ▶ Hot surfaces and molten metal
- ▶ Chemical splashes
- ▶ Electrostatic buildup

Leg protection PPE include safety boots, foot guards, knee pads and leggings.



Boots



Foot guards



Knee guard



Leggings



2.8 Body Protection

Persons working in low visibility and high movement zones need to wear high visibility vests to avoid collisions



High Visibility Vest



PPE

2.9 Fall Protection

Falls are a risk when working at heights and in confined spaces.

Fall arrestors PPE prevent such falls.



Lanyard body
harness



3. Specialized PPE

- ▶ Handling Hazardous Chemical & Biological materials
- ▶ High Voltage Electrical Work
- ▶ Working in Extreme Heat conditions and in Fire Fighting

